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- CLASS : SYIT
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```
import pandas as pd
```

5a. Write a python code to demonstrate importing pandas libraries and create data frame object.

```
import pandas as pd

data = {
    "name": ["Sharu", "Mahesh", "Jay"],
    "age": [50, 80, 20],
    "qualified": [True, False, False]
}
df = pd.DataFrame(data)
print(df)
print("JUELI GAWADE S008")
```

```
      name  age  qualified
0  Sharu   50         True
1  Mahesh   80         False
2    Jay   20         False
JUELI GAWADE S008
```

5b. Write a python code to show statistical information on given data set.

```
import statistics
```

MEAN DEVIATION

```
print(statistics.mean([1, 3, 5, 7, 9, 11, 13]))
print(statistics.mean([1, 3, 5, 7, 9, 11]))
print(statistics.mean([-11, 5.5, -3.4, 7.1, -9, 22]))
print("JUELI GAWADE S008")
```

```
7
6
1.8666666666666667
JUELI GAWADE S008
```

MEDIAN

```
print(statistics.median([1, 3, 5, 7, 9, 11, 13]))
print(statistics.median([1, 3, 5, 7, 9, 11]))
print(statistics.median([-11, 5.5, -3.4, 7.1, -9, 22]))
print("JUELI GAWADE S008")
```

```
7
6.0
1.05
JUELI GAWADE S008
```

MODE

```
print(statistics.mode([1, 3, 3, 3, 5, 7, 7, 9, 11]))
print(statistics.mode([1, 1, 3, -5, 7, -9, 11]))
print(statistics.mode(['red', 'green', 'blue', 'red']))
print("JUELI GAWADE S008")
```

```
3
1
red
JUELI GAWADE S008
```

STANDARD DEVIATION

```
print(statistics.stdev([1, 3, 5, 7, 9, 11]))
print(statistics.stdev([2, 2.5, 1.25, 3.1, 1.75, 2.8]))
print(statistics.stdev([-11, 5.5, -3.4, 7.1]))
print(statistics.stdev([1, 30, 50, 100]))
print("JUELI GAWADE S008")
```

```
3.7416573867739413
0.6925797186365383
8.414471660973927
41.67633221226007
JUELI GAWADE S008
```

VARIANCE

```
print(statistics.variance([1, 3, 5, 7, 9, 11]))
print(statistics.variance([2, 2.5, 1.25, 3.1, 1.75, 2.8]))
print(statistics.variance([-11, 5.5, -3.4, 7.1]))
print(statistics.variance([1, 30, 50, 100]))
print("JUELI GAWADE S008")
```

```
14
0.47966666666666663
70.80333333333333
1736.9166666666667
JUELI GAWADE S008
```

5c. Write a python code to create pandas series from dictionaries.

SERIES

```
# create a dictionary
dictionary = {'A': 10, 'B': 20, 'C': 30}

# create a series
series = pd.Series(dictionary)

print(series)
print("JUELI GAWADE S008")
```

```
A    10
B    20
C    30
dtype: int64
JUELI GAWADE S008
```

INDEX

```
# create a dictionary
dictionary = {'A': 50, 'B': 10, 'C': 80}

# create a series
series = pd.Series(dictionary, index=['B', 'C', 'A'])

print(series)
print("JUELI GAWADE S008")
```

```
B    10
C    80
A    50
dtype: int64
JUELI GAWADE S008
```

5d. Write a python code to demonstrate filter pandas series with Boolean arrays.

```
# dictionary of lists
dict = {'name': ["aparna", "pankaj", "sudhir", "Geeku"],
        'degree': ["MBA", "BCA", "M.Tech", "MBA"],
        'score': [90, 40, 80, 98]}

df = pd.DataFrame(dict, index = [True, False, True, False])

print(df)
print("JUELI GAWADE S008")
```

```
      name degree  score
True  aparna   MBA    90
False pankaj   BCA    40
True  sudhir  M.Tech   80
```

False	Geeku	MBA	98
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